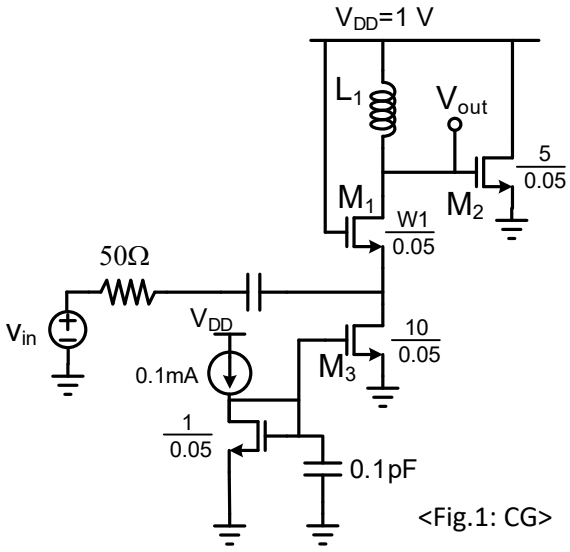
ECE6420 HW5

Use the active device model used in the Cadence tutorial.

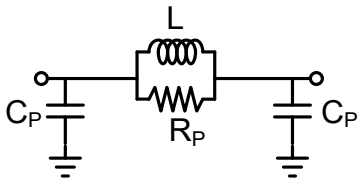
(a) Consider the common gate circuit shown in Fig.1, where M_2 represents a typical load capacitance for now. Assuming $\lambda = 0$, find the width of M1 such that $g_{m1} + g_{mb1} = (50\Omega)^{-1}$.

(b) Use simulations to determine the value of L_1 for the resonance at the output at 5.8 GHz. The inductor must be modeled as shown in Fig.2, where Rp is chosen to give a Q of 8 at the frequency of interest and Cp=10fF for every nanohenry of inductance.



<Fig.1: CG>

(c) Now find the input resistance (i.e., the real part of the input impedance) of Fig.1 with the modeled inductor in (b) by simulations and explain why it is not equal to 50Ω. Adjust the width of M1 to obtain a 50Ω input resistance, and tune the inductor to resonate at 5.8GHz again.

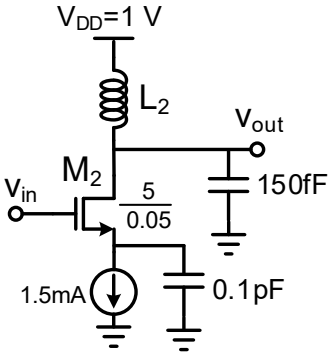


<Fig.2: Realistic inductor model>

(d) Using the shortcut method (two tone test) to compute the IIP3 and voltage gain of the circuit at 5.8 GHz.

(e) Determine the IIP3 and voltage gain of the stage shown in Fig.3 at 5.8 GHz. The role and modeling of L_2 are similar to those of L_1.

(f) Now replace M2 in Fig.1 with the circuit in Fig.3, and find the overall voltage gain and IIP3 of the cascaded structure. How closely do these results agree with those obtained from parts (d) and (e) and Eq. (2.67) in the text?



<Fig.3: 2nd stage>

(g) Does the input resistance of the first stage change when the second stage is added? Why?